Smart Pregnancy Belt – Design Overview

Original infographic supplied by the user (preserved in full)

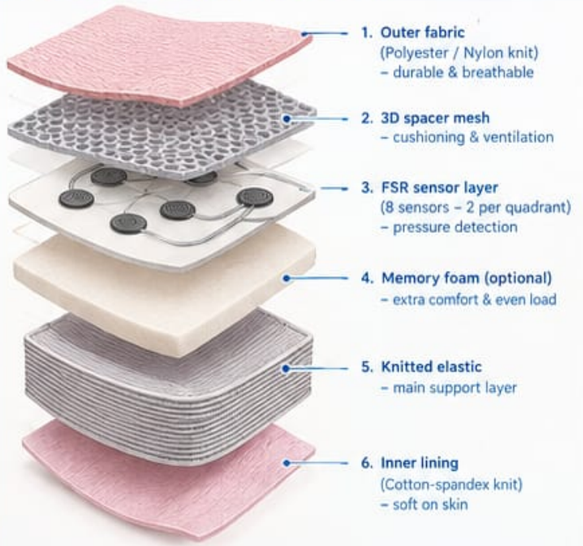
Smart Pregnancy Belt – Design Overview

Comfort • Pressure Monitoring • Automatic Fit Adjustment • Safe for Your & Your Baby

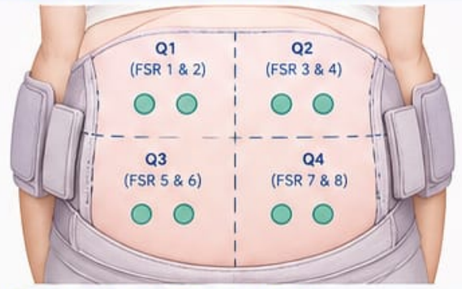
8 FSR sensors – 2 in each quadrant
Detects uneven pressure and incorrect fit, helping to redistribute load and adjust the belt when needed.



Layered Structure (Cross-section View)

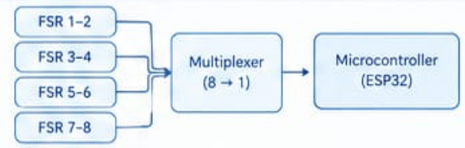


Sensor Placement (Top View – Belly)



Each quadrant has 2 FSR sensors placed symmetrically to detect pressure distribution.

Sensor Wiring (Simplified)



All 8 sensors connect to a multiplexer to save GPIO pins. Data is processed in the ESP32.

Where Each Material is Used



1 Outer fabric (Polyester / Nylon knit)
- Used for outer surface (durable, breathable, washable)

2 3D spacer mesh
- Used in abdominal panel for cushioning & airflow

3 FSR sensor layer (8 sensors)
- Placed between spacer mesh and foam

4 Memory foam (optional)
- Adds comfort and reduces sharp pressure points

5 Knitted elastic
- Used in support band and side straps

6 Inner lining (Cotton-spandex knit)
- Touches skin for comfort and prevents irritation

Material Selection (Recommended)

Part / Layer	Material	Why it's suitable
Inner lining (skin contact)	Cotton-spandex knit (95% cotton + 5% spandex)	Soft, breathable, prevents irritation
Outer fabric	Polyester / Nylon knit	Durable, lightweight, washable
Cushioning layer	3D spacer mesh (Polyester)	Distributes pressure, allows airflow
Support band	Knitted elastic (Nylon + spandex)	Flexibility + support without tight compression
Side straps	Nylon / Polyester webbing	Strong, durable, resists stretching
Fasteners	Hook-and-loop (Velcro)	Adjustable, easy to use
Sensor layer	Flexible FSR (8 units)	Pressure detection
Wiring	Flexible insulated wire	Safe, lightweight
Electronics cover	Medical-grade silicone or TPU	Protects electronics, water-resistant

Bill of Materials (BOM) – Prototype

Component	Specification / Example	Quantity
Cotton-spandex fabric (inner)	95/5 knit, 200 gsm	0.5 – 1 m
Polyester/nylon fabric (outer)	Breathable knit, 180 gsm	0.5 – 1 m
3D spacer mesh	3 mm thickness	1 panel
Memory foam (optional)	PU foam, 5 mm	1 panel
Knitted elastic	Nylon + spandex, 8–15 cm width	1–2 m
Nylon/polyester webbing	For side straps	1–2 m
Hook-and-loop (Velcro)	2–4 cm width	2–4 pieces
FSR sensors	Thin flexible FSR	8
IMU sensor	6-axis (e.g., MPU6050)	1
Strain sensor (optional)	Flexible strain sensor	2
Microcontroller	ESP32 Dev board	1
Vibration motor (alert)	Coin vibration motor	1
Battery	LiPo 3.7V (with protection)	1
Wires & connectors	Flexible, insulated	As needed
Electronics enclosure	Silicone/TPU case	1

Design Highlights for Your Prototype

8 FSR sensors
2 per quadrant for full belly coverage

Even pressure distribution
Detects concentrated pressure and misalignment

Smart adjustment
Alerts wearer or adjusts straps (safe limits)

Comfort & safety
Breathable, soft, skin-friendly materials

Important Note

This is a prototype design recommendation. For actual use in pregnancy, get the design reviewed by a medical professional (OB/GYN or physiotherapist) to ensure safety, comfort and appropriate support levels.

Optional Add-ons (Future Versions)

- Wireless connectivity (Bluetooth/Wi-Fi)
- Mobile app for real-time data
- Auto-adjustment using small motors (only with safety limits)
- Water-resistant electronics cover

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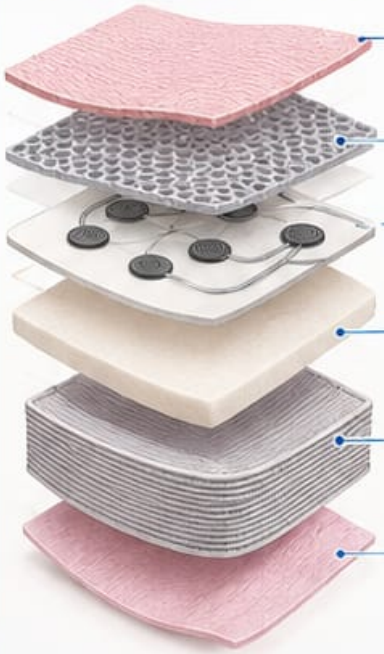
8 FSR sensors – 2 in

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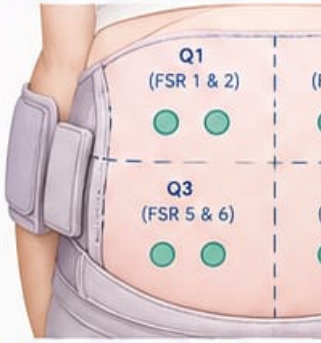
al belt look (front view)

Layered Structure (Cross-section View)



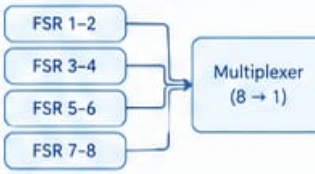
1. **Outer fabric** (Polyester / Nylon knit) – durable & breathable
2. **3D spacer mesh** – cushioning & ventilation
3. **FSR sensor layer** (8 sensors – 2 per quadrant) – pressure detection
4. **Memory foam (optional)** – extra comfort & even load
5. **Knitted elastic** – main support layer
6. **Inner lining** (Cotton-spandex knit) – soft on skin

Sensor Placement (Top Vie



Each quadrant has 2 FSR sensors p symmetrically to detect pressure d

Sensor Wiring (Simplified)



All 8 sensors connect to a multiplexe GPIO pins. Data is processed in the E

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1. **Outer fabric** (Polyester / Nylon knit)
   - Used for outer surface (durable, breathable, washable)
2. **3D spacer mesh**
   - Used in abdominal panel for cushioning & airflow
3. **FSR sensor layer (8 sensors)**
   - Placed between spacer mesh and foam
4. **Memory foam (optional)**
   - Adds comfort and reduces sharp pressure points
5. **Knitted elastic**
   - Used in support band and side straps
6. **Inner lining** (Cotton-spandex knit)
   - Touches skin for comfort and prevents irritation

Material Selection (Recommended)

Part / Layer	Material	Why it's suitable
Inner lining (skin contact)	Cotton-spandex knit (95% cotton + 5% spandex)	Soft, breathable, prevents irritation
Outer fabric	Polyester / Nylon knit	Durable, lightweight, washable
Cushioning layer	3D spacer mesh (Polyester)	Distributes pressure, allows airflow
Support band	Knitted elastic (Nylon + spandex)	Flexibility + support without tight compression
Side straps	Nylon / Polyester webbing	Strong, durable, resists stretching
Fasteners	Hook-and-loop (Velcro)	Adjustable, easy to use
Sensor layer	Flexible FSR (8 units)	Pressure detection
Wiring	Flexible insulated wire	Safe, lightweight
Electronics cover	Medical-grade silicone or TPU	Protects electronics, water-resistant

Bill of Materials (BOM) – Prototyp

Component	Specification /
Cotton-spandex fabric (inner)	95/5 knit, 200 gs
Polyester/nylon fabric (outer)	Breathable knit,
3D spacer mesh	3 mm thickness
Memory foam (optional)	PU foam, 5 mm
Knitted elastic	Nylon + spandex,
Nylon/polyester webbing	For side straps
Hook-and-loop (Velcro)	2–4 cm width
FSR sensors	Thin flexible FSR
IMU sensor	6-axis (e.g., MPU
Strain sensor (optional)	Flexible strain se
Microcontroller	ESP32 Dev board
Vibration motor (alert)	Coin vibration mo
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Electronics enclosure	Silicone/TPU case

r Your Prototype

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Even pressure distribution
Detects concentrated pressure and misalignment

Comfort & safety
Breathable, soft, skin-friendly materials

Important Note

This is a prototype design recommendation. For actual use in pregnancy, get the design reviewed by a medical professional (OB/GYN or physiotherapist) to ensure safety, comfort and appropriate support levels.

Optional Add-ons (Future Versions)

- Wireless connectivity (Bluetooth/Wi-Fi)
- Mobile app for real-time data
- Auto-adjust using small (only with sa

Detailed Written Explanation

This section transcribes and explains the components, materials, sensor arrangement, electronics, prototype bill of materials, and design notes shown in the supplied infographic. The infographic is a conceptual prototype proposal, not a clinically validated medical device.

1. Overall design purpose

The concept is a **Smart Pregnancy Belt** intended to combine four goals: **comfort**, **pressure monitoring**, **automatic fit adjustment**, and **support intended to be safe for the wearer and baby**. The belt uses flexible textile layers for abdominal support, force-sensitive resistor (FSR) sensors to estimate local pressure distribution, and an electronics module to read the sensors. The concept aims to identify uneven pressure or an incorrect fit and then alert the wearer or, in a future version, adjust the belt within defined safety limits.

Important interpretation: FSR sensors measure force-related changes at their sensing areas; they do not automatically provide clinically accurate pressure values without calibration and mechanical validation. The proposed automatic adjustment is a future design function and should not be assumed safe or implemented for pregnancy use without qualified clinical and engineering review.

2. Final belt appearance and visible components

The front-view illustration shows a wide abdominal support panel worn around the lower abdomen. The pictured belt includes the following labelled elements:

• **Wide adjustable side straps (with Velcro / hook-and-loop):** These allow the wearer or caregiver to change the fit. The strap system should distribute load without creating tight local bands.

• **Electronics module (under cover):** A small housing mounted near the side of the belt. It is intended to contain the microcontroller and associated circuitry. The cover should protect the electronics from incidental contact and moisture while remaining comfortable.

• **Abdominal support panel (with sensors and cushioning):** The main front panel supports the abdomen and contains the sensing and cushioning layers.

• **Breathable fabric (inner layer):** The skin-facing textile should feel soft, allow moisture and heat to escape, and reduce rubbing or irritation.

• **Final belt look (front view):** The illustration presents a wrap-around support belt with a central panel and adjustable side closures.

3. Layered structure (cross-section view)

The infographic depicts six layers stacked from the outside of the belt toward the skin. The sequence matters because the outer textile protects the assembly, the spacer mesh and foam provide cushioning, the FSR layer senses loading, the elastic layer provides support, and the lining forms the skin-contact surface.

Layer	Material / description	Role in the design
1. Outer fabric	Polyester / nylon knit; durable and breathable	Outer protective surface. Intended to tolerate ordinary handling and washing appropriate to the selected textile and construction.
2. 3D spacer mesh	Open, three-dimensional mesh	Cushioning and ventilation. Its open structure can permit airflow and help distribute contact loads.
3. FSR sensor layer	8 sensors total; 2 sensors per quadrant	Pressure-related sensing. Sensor placement allows comparisons between regions of the abdominal panel.
4. Memory foam (optional)	Soft cushioning foam	Extra comfort and more even loading; may reduce sharp pressure points. Its thickness and stiffness can also affect sensor response.
5. Knitted elastic	Nylon + spandex	Main support layer. Intended to stretch and provide support without excessive compression.
6. Inner lining	Cotton-spandex knit	Soft skin-contact layer, intended to improve comfort and reduce irritation.

Construction note: The diagram is schematic. The exact stack order, stitching, sensor encapsulation, foam placement, and load path must be finalized through prototype tests. Placing foam or mesh between the body and a sensor can change the sensor's response, so the complete assembled belt—not only the bare sensor—must be calibrated.

4. Sensor placement: top view of the belly

The abdominal panel is divided into four labelled quadrants. Each quadrant has two FSR sensors, for a total of eight sensors:

• **Q1:** FSR 1 and FSR 2.

• **Q2:** FSR 3 and FSR 4.

• **Q3:** FSR 5 and FSR 6.

• **Q4:** FSR 7 and FSR 8.

The infographic places the two sensors in each quadrant symmetrically. Comparing readings between left/right and upper/lower regions can help identify uneven loading, asymmetry, or a change in fit. The sensor readings should be interpreted as relative or calibrated measurements; body movement, fabric tension, sensor drift, posture, and the belt's construction can all affect readings.

5. Sensor wiring and electronics (simplified)

The diagram groups the eight sensors into four pairs: FSR 1–2, FSR 3–4, FSR 5–6, and FSR 7–8. These sensor signals are routed to a **multiplexer (8-to-1)**, which selects one sensor channel at a time and passes that channel to the microcontroller. The microcontroller is identified as an **ESP32**.

The infographic explains that all eight sensors connect to a multiplexer to save GPIO pins, and that data is processed in the ESP32. In a practical circuit, the sensor interface may require suitable voltage-divider or conditioning circuits, stable power, multiplexer select-line control, analogue-input compatibility, filtering, and calibration. The exact circuit depends on the chosen FSR model and ESP32 board.

Possible data flow: mechanical load → FSR resistance changes → sensor interface / multiplexer selects a channel → ESP32 reads the signal → software filters and compares readings → warning or display output. The infographic also lists a vibration motor as an alert component.

6. Where each material is used

Number	Material / component	Use stated in the infographic
1	Outer fabric (polyester / nylon knit)	Used for the outer surface; described as durable, breathable, and washable.
2	3D spacer mesh	Used in the abdominal panel for cushioning and airflow.
3	FSR sensor layer (8 sensors)	Placed between spacer mesh and foam, as shown in the layer illustration.
4	Memory foam (optional)	Adds comfort and reduces sharp pressure points.
5	Knitted elastic	Used in the support band and side straps.
6	Inner lining (cotton-spandex knit)	Touches the skin for comfort and to help prevent irritation.

7. Recommended material selection

Part / layer	Material	Why it is suitable (as stated / intended)
Inner lining (skin contact)	Cotton-spandex knit (95% cotton + 5% spandex)	Soft and breathable; intended to help prevent irritation.
Outer fabric	Polyester / nylon knit	Durable, lightweight, and washable.
Cushioning layer	3D spacer mesh (polyester)	Distributes pressure and allows airflow.
Support band	Knitted elastic (nylon + spandex)	Flexibility and support without tight compression.
Side straps	Nylon / polyester webbing	Strong and durable; resists stretching.

Part / layer	Material	Why it is suitable (as stated / intended)
Fasteners	Hook-and-loop (Velcro)	Adjustable and easy to use.
Sensor layer	Flexible FSR (8 units)	Pressure-related detection.
Wiring	Flexible insulated wire	Intended to be safe and lightweight.
Electronics cover	Medical-grade silicone or TPU	Protects electronics and is intended to be water-resistant.

8. Bill of materials (BOM) for the prototype

The table below reproduces the example specifications and quantities shown in the image. These are preliminary prototype estimates, not a finalized manufacturing or purchasing list.

Component	Specification / example	Quantity
Cotton-spandex fabric (inner)	95/5 knit, 200 gsm	0.5–1 m
Polyester/nylon fabric (outer)	Breathable knit, 180 gsm	0.5–1 m
3D spacer mesh	3 mm thickness	1 panel
Memory foam (optional)	PU foam, 5 mm	1 panel
Knitted elastic	Nylon + spandex, 8–15 cm width	1–2 m
Nylon/polyester webbing	For side straps	1–2 m
Hook-and-loop (Velcro)	2–4 cm width	2–4 pieces
FSR sensors	Thin flexible FSR	8
IMU sensor	6-axis (e.g., MPU6050)	1
Strain sensor (optional)	Flexible strain sensor	2
Microcontroller	ESP32 Dev board	1
Vibration motor (alert)	Coin vibration motor	1
Battery	LiPo 3.7 V (with protection)	1
Wires & connectors	Flexible, insulated	As needed
Electronics enclosure	Silicone/TPU case	1

9. Design highlights for the prototype

8 FSR sensors: Two sensors in each of four quadrants provide coverage across the abdominal panel.

Even pressure distribution: The sensor array is intended to help detect concentrated pressure and misalignment.

Smart adjustment: The concept describes alerts to the wearer or strap adjustment within defined safety limits.

Comfort and safety: The intended materials are breathable, soft, and skin-friendly.

10. Important note from the infographic

The image explicitly states that this is a prototype design recommendation. For actual use during pregnancy, the design should be reviewed by a medical professional (OB/GYN or physiotherapist) to ensure safety, comfort, and appropriate support levels.

This caution is especially important for any automatic tightening or fit-adjustment feature. A prototype should begin with passive monitoring and user alerts; any motor-driven adjustment should have conservative limits, a manual release, a fail-safe state, and clinical oversight before any real-world use.

11. Optional add-ons for future versions

Wireless connectivity: Bluetooth / Wi-Fi communication to send readings to another device.

Mobile app: A phone interface for real-time data and trends.

Auto-adjustment using small motors: Proposed only within safety limits; requires rigorous mechanical and clinical validation.

Water-resistant electronics cover: A cover intended to protect the electronic module from incidental moisture.

12. Suggested prototype development and verification plan

• **Textile and comfort tests:** check seams, edges, stretch, breathability, skin-contact comfort, and whether the belt slips during movement.

• **Sensor calibration:** calibrate each FSR in the complete layered assembly using known loads; document repeatability, hysteresis, drift, and sensor-to-sensor variation.

• **Fit and placement checks:** verify that sensor locations remain consistent and that readings are not dominated by wrinkles, seams, or electronics housing pressure.

• **Electronics checks:** confirm safe battery protection, stable voltage, electrical insulation, connector strain relief, and reliable sampling of all eight channels.

• **Software checks:** record raw readings, filter noise, detect disconnected or saturated sensors, and set warning thresholds only after suitable testing.

• **Safety controls:** avoid automatic tightening on a pregnant wearer until a qualified clinical and engineering review has defined and validated allowable forces, fail-safe behaviour, and a manual release.

• **Documentation:** keep a record of materials, dimensions, sensor positions, calibration curves, test conditions, limitations, and revisions.

13. Terms explained in simple language

FSR (Force-Sensitive Resistor): A sensor whose electrical resistance changes when force is applied. The electronics convert this change into a reading that can be calibrated against applied load.

Quadrant: One of four regions made by dividing the panel into upper-left, upper-right, lower-left, and lower-right sections.

3D spacer mesh: A thick, porous textile with an open structure that creates space for cushioning and airflow.

Memory foam: A foam that deforms under load and recovers gradually; in this design it is optional cushioning.

Knitted elastic: A stretchable textile band used to provide support while allowing some movement.

Multiplexer (8-to-1): An electronic selector that routes one of eight input signals to a shared output, reducing the number of microcontroller input channels needed.

ESP32: A family of microcontroller boards commonly used to read sensors and support wireless communication.

IMU: An inertial measurement unit; a 6-axis unit typically combines accelerometer and gyroscope measurements to estimate movement and orientation.

Strain sensor: A sensor that changes its electrical output when stretched or deformed; listed as optional in this concept.

LiPo battery: A lithium-polymer rechargeable battery. It needs appropriate charging, protection, and enclosure design.

TPU: Thermoplastic polyurethane, a flexible polymer that can be used in protective covers.

BOM: Bill of materials: a list of the parts, example specifications, and quantities needed to build a prototype.

14. Limits and assumptions to keep in mind

• The image does not specify exact belt dimensions, pattern pieces, sensor model numbers, circuit schematics, firmware, calibration curves, or validated pressure thresholds.

• The BOM quantities are rough estimates and may change with body-size range, sewing pattern, prototypes, sensor dimensions, and connector routing.

• FSR readings are affected by the mechanical stack and may be nonlinear, so the system should not be described as measuring medically accurate pressure until validated.

• The concept does not establish that the belt prevents pregnancy complications or guarantees safety for the fetus or wearer.

• The optional motorized adjustment, mobile app, wireless features, IMU, and strain sensors need separate design and verification work.

End of detailed notes. The original infographic is included as the first page of this PDF so the visual reference and the expanded written explanation remain together.